

PANJABUNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. Under the Panjab University Act VII of 1947 –enacted by the Govt. of India)

SYLLABI

Course Outline for
B.Voc (Medical Lab Technology)

As per NEP 2020

w.e.f. session 2024-25

Course Outline for B.Voc Medical Lab Technology
as per NEP 2020 w.e.f. session 2024-25

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
MLT 101	Core Discipline	Basics of Human Anatomy	2	2	4	40	10	40	10	100
MLT 102	Core Discipline	Fundamentals of Microbiology	2	2	4	40	10	40	10	100
MLT 103	Core Discipline	Introduction to Hematology	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 101	Multidisciplinary Course	Fundamentals of Information Technology[1]	3	-	3	60	15	-	-	75
SEC I	Skill Enhancement Course	SEC Options[2]			3					75
VAC I	Value-Added Course	VAC Options[3]			2					50
PW I	Project Work	Project Work-I & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
MLT 201	Core Discipline	Basics of Physiology	2	2	4	40	10	40	10	100
MLT 202	Core Discipline	Basics of Biochemistry	2	2	4	40	10	40	10	100
MLT 203	Core Discipline	Analytical Laboratory Testing Process-I	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development[1]	3	-	3	60	15	-	-	75
SEC II	Skill Enhancement Course	SEC Options[2]			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role – Medical Lab Technician

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3weeks of the start of 3rdsemester.

** Minimum Credits required to get Certificate Course in Medical lab technology= 48 + 4 #.

#Exit 1: Certificate Course-Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

ESSENTIAL COMPONENTS OF COURSE OF
B.VOC Medical Lab Technology
(SEMESTERS I & II)

PROGRAMME OUTCOMES

PO1. Demonstrate technical skills, social behavior and professional awareness for functioning effectively as a laboratory technician.

PO2.Recognize the impact of laboratory tests in a global and environmental context.

PO3Apply knowledge and technical skills associated with medical laboratory technology for delivering quality clinical investigations support.

PO4: Perform routine clinical laboratory procedures within acceptable quality control parameters in hematology, biochemistry, immunohematology and microbiology.

PO5: Demonstrate technical skills, social behavior and professional awareness for functioning effectively as a laboratory technician.

PO6: Operate and maintain laboratory equipments utilizing appropriate quality control and safety procedures.

PO7: Communicate effectively by oral, written and graphical means.

PO8: Function as a leader / team member in diverse professional and industrial research areas.

PO9: Function in an ethical and professional manner without bias against any ethnicity, race, religion, caste or gender.

PO10: Practice professional and ethical responsibilities with high degree of credibility, integrity and social concern.

Semester I

	COURSE TITLE	BASICS OF HUMAN ANATOMY
1.	**CourseCode**	MLT 101
2.	**Typologyof Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100 Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Medical Lab Technician
7.	**Course Objectives**	Basic understanding of organization of body cells, tissues, organs, organ systems, and glands in human body.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1.Understand the human body and increase scientific knowledge about different anatomical terms and positions of human body. CO2. Get the theoretical as well as practical knowledge regarding basic organization of human body so that they can pursue their studies in Medical Lab Technology CO3.Learn basic landmarks and position of various organs along with their structure.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT I

Basic unit of body - Cell
The anatomical organization of body cells, tissues, organs, organ systems, Membranes and glands in human body.
Introduction to different types of tissues: Epithelial Tissue, Muscular Tissue, Nervous Tissue
Different types of organ systems.
Brief Introduction of different types of body fluids, secretions and excretions
Skeletal system: bones, joints and muscles.

UNIT II

Digestive Organs:
- Tongue
- Teeth
- Oral cavity
- Pharynx
- Oesophagus
- Stomach
- Small intestine
- Large intestine
- Liver, Pancreas and Spleen

UNIT III

Respiratory Organs:
- Nasopharynx
- Oropharynx
- Larynx
- Trachea
- Bronchi
- Lungs
- Thoracic cavity
- Pleura and Pleural cavity

Circulation System: Structure of Heart and Brief introduction of main blood vessels.

UNIT IV

Reproductive Organs Male and Female Gonads
Nervous system and Sense organs: Brief Introduction of Central Nervous System and Peripheral Nervous System
Anatomy of Brain , Spinal Cord, Nerves, Eye, Ear, Olfactory Receptors, Gustatory Receptors
Excretory Organs:
- Cortex and medulla of Kidney
- Ureter
- Urinary Bladder

Urethra (male and female)

Practical Work:

To study of various organ systems through Charts and Models

12. Suggested Resources for Reading

1.	Anatomy & Physiology:	Ross and Wilson
2.	Anatomy and Physiology:	N Murgesh
3.	Anatomy and Physiology for nurses:	Evelyn Pearce
4.	Anatomy and Physiology for nurses:	Sears
5.	Anatomy and Physiology for nurses:	Pearson
6.	Human Anatomy:	Harie R. Berasari

Note: Latest edition of text books may be used.

	COURSE TITLE	FUNDAMENTALS OF MICROBIOLOGY
1.	**CourseCode**	MLT 102
2.	**Typologyof Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (80TH/PR+20IA) i) The exam of this course will be of 3 hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Medical Lab Technician
7.	**Course Objectives**	To give an overview of various aspects of General microbiology
8.	**Course Outcomes**	CO1. On successful completion of the course, the students will be able to understand the various aspects of general microbiology
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT I

Introduction and brief History, development, scope and applications of Microbiology.
Nature of Microbial World: Prokaryotes and eukaryotes, growth pattern in microbes
Morphology and Structure of Microorganisms
Growth and nutrition of microbes
Safety measures in microbiology.

UNIT II

Morphology, General characteristics & fine structure of bacteria, fungi, actinomycete and algae
Classification of bacteria.
Organization of cell wall, cell membrane, flagella and capsules in bacteria.
Morphogenesis in bacteria, formation of spores and cysts**.**

UNIT III

General characteristics actinomycete and algae
Morphology, General characteristics and classification of pathogenic fungi.
Classification and general properties of Viruses
Morphology, Pathogenicity and laboratory diagnosis of human viruses.
Bacteriophages: Morphology, multiplication, detection and enumeration**.**

UNIT IV

Methods of Microbiology isolation of pure cultures, theory and practice of sterilization.
Staining of microbes, Theory of Gram staining.
Preparation, uses and standardization of culture media.
Principles of staining methods and preparation of reagents.
Principles and methods of sterilization.
Uses and mode of action antiseptics and disinfectants.

Practical Work:

1. Use of microscope in examination of unstained bacteria, fungi, algae, parasites and stained cell preparations including simple staining,
2. Gram's staining, acid fast staining, capsule staining, spore staining using prokaryotic and eukaryotic cells, hanging drop preparation.
3. Preparation of culture media, spread plates, pour plates, selective media, differential media
4. Separation of pure cultures

12. Suggested Resources for Reading

1. Mims' Medical Microbiology Richard Goering, Hazel Dockrell, Mark Zuckerman, Ivan M. Roitt, Professor Peter L. Chiodini Publisher Elsevier Health Sciences
2. Ananthnaryans and Paniker's Text book of Microbiology edited by CKJ Paniker 7th edition, Publisher Orient Longman
3. Roitt's Essential immunology Delves, Peter J., Martin, Seamus J. Burton, Dennis R. Roitt, Ivan M Ananthanarayan and Paniker's Textbookof microbiology Kapil, Arti ed
4. Microbiology: An Introduction,12th edition Gerard J. Tortora Berdell R. Funke and Christine L. Case
5. Parasitology Chatterjee K.D.
6. Microbiology Pelczar, Michal J and Others

7. Medical microbiology Greenwood David and Other
8. Ananthanarayan and Panikar's text book of Microbiology Arti Kapil
9. Immunology Male David and Other
10. Mackie and Mc Cartney practical Medical Microbiology Collee J.G and Other
11. Bailey and Scott's Diagnostic Microbiology,13th edition Patricia Tille
12. Sherris Medical Microbiology, 6th edition Pottinger, L. Barth Reller and Charles R. Sterling
13. Stephenson Calculations for Molecular Biology
14. Gerald Karp Cell Molecular Biology
15. Medical Informatics Mohan Bansal

Note: Latest edition of text books may be used.

	COURSE TITLE	INTRODUCTION TO HEMATOLOGY
1.	**CourseCode**	MLT 103
2.	**Typologyof Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Medical Lab Technician
7.	**Course Objectives**	To gain understanding of blood and component of blood. To gain knowledge of hematological diseases and hematological investigations.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1 To get understanding of blood and its components as well as hematological techniques and diseases CO2. Differentiate various laboratory test findings with their associated clinical conditions. CO3. Be able to distinguish the developmental stages of blood cells. It will also cover Bone marrow examination. CO4. Identify the various skills necessary to perform blood counts, evaluate blood elements and report within the stated limits of accuracy.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT I

Introduction to hematology and laboratory organization Composition and functions of blood and lymph.
Detailed study of Haemoglobin and its functions of hemoglobin
Blood groups including Rh. Factor
Detailed study of Reticulocytes
Formation of blood. Morphology of normal blood cells and their identifications
Hemostasis, Mechanism of blood coagulation. Fibrinolysis**.**

UNIT II

Various anticoagulants, their uses, mode of action and their merits and demerits. Normal and absolute in hematology. Quality assurance in hematology.

UNIT III

Descriptive study of RBC abnormalities, Disorders related to RBC Normal white cell count & physiological variation Normal white cell count & physiological variation Hematological Diseases: Anemia and various types of anemia, Thalassemia, Polycythemia, hemolytic disease of new born, multiple myeloma, parasitic infections of blood Leukemia: definition and classification (General & FAB).

UNIT IV

Laboratory Investigation & Bleeding Disorders
Laboratory preparation for coagulation tests, Routine coagulation tests, prothrombin time, plasma recalcification time, partial thromboplastin time, activated partial thromboplastin time, thrombin time, Laboratory diagnosis of bleeding disorders.

Practical Work:

1. Cleaning of Laboratory glassware in hematology
2. Clinical significance, specimen collection, laboratory investigation & preservation of blood for various hematological investigations.
3. Preparation of blood smear.
4. Haemocytometery, procedures for cell counts-visual as well as electronic
5. Total leukocyte counts and Differential leukocyte count.
6. Determination of total erythrocyte (RBC) count and platelet counts. Errors involved and mean to minimize such errors.
7. Haemoglobinometery, various methods of estimation of Hb, errors involved and standardization of instrument for adaptation for Hb estimation.
8. Romanowsky dyes, preparation and staining procedures of blood smears.
9. Laboratory tests for assessing bleeding disorders

12. Suggested Resources for Reading

1. Atlas of haematology (5/e) G.A. McDonald
2. Clinical Haematology Christopher A. Ludlam
3. Practical Haematology J.B. Dacie
4. Practical Haematology (8/e) S ir John
5. Haematology (International edition) Emmanuel C. Besa
6. Haematology (Pathophysiological basis for clinical practice (3/e) Stephen M. Robinson
7. Haematology for students Practitioners Ramnik Sood
8. Hand book of Medical Laboratory Technology (2/e) V.H. Talib

Note: Latest edition of text books may be used.

Semester II

	COURSE TITLE	BASICS OF PHYSIOLOGY
1.	**Course Code**	MLT 201
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Medical Lab Technician
7.	**Course Objectives**	To understand the basics of physiology of different organ system of body. .
8.	**Course Outcomes**	CO1. On successful completion of the course, the students will be able to the understand basics of physiology of different organ system of body to help students better understanding medical lab sciences and lab tests.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT I

Nutrition- Different types of Nutrients and Vitamins
Integumentary system: Structure of Skin and its functions.
Digestive system: Physiology of digestion of food and its absorption**.**

UNIT II

Lymphatic system: Different types of body fluids and their functions
Spleen, lymph node and R.E. system
Excretory System: Urine formation, osmoregulation and counter current mechanism.
Cardiovascular system: Origin and regulation of heart beat, cardiac cycle, electrocardiogram, cardiac output, blood pressure and micro-circulation.

UNIT III

Respiratory system- Transport of O2 and CO2, Oxygen dissociation curve of haemoglobin, Bohr effect, chloride (-) shift, Haldane effect
Muscular System: Ultra structure and physiological basis of skeletal muscle contractions.
Reproductive System- Brief Introduction to Female Reproductive System and Male Reproductive System
Fertilization and Gametogenesis**.**

UNIT IV

Neural Physiology- Structure of neuron, resting membrane potential, origin and propagation of impulse along the axon, synapse and myoneural junction.
Endocrine System- Structure, and functions of hormones of thyroid, parathyroid, adrenal, pineal, hypothalamus, pituitary, pancreas, gonads, thymus.
Hormones of alimentary canal and kidney.

Practical Work:

1. To Study of body systems mentioned in theory through Charts.
2. To demonstrate the separate the plasma and serum from given blood sample.
3. To study of various endocrine glands through permanent slides /Charts.

12. Suggested Resources for Reading

1. Textbook of Medical Physiology Guyton and Hall
2. Anatomy & Physiology Ross and Wilson
3. Anatomy and Physiology N Murgesh
4. Anatomy and Physiology for nurses Evelyn Pearce
5. Anatomy and Physiology for nurses Sears
6. Anatomy and Physiology for nurses Pearson
7. Anatomy and Physiology: Understanding the Human Body Clark
8. Physiology & Health Education Gandhi & Goel
9. Endocrinology Headley
10. Human Physiology Andrew Davis
11. Manual of Endocrinology and Metabolism Norman Levin

Note: Latest edition of text books may be used.

	COURSE TITLE	BASICS OF BIOCHEMISTRY
1.	**CourseCode**	MLT 202
2.	**Typologyof Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Medical Lab Technician
7.	**Course Objectives**	To gain elementary knowledge of biochemistry.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1.Understand the biochemistry of Carbohydrates, Proteins and Fats and their clinical significance. CO2. Evaluate problems & results involving Acid-base and pH chemistry, atomic structure, Solution and Dialysis CO3. Evaluate principles of Biochemistry of Enzymes and Nucleic acids. CO4. Learn about Radioisotopes and their uses in Biochemistry. CO5. Analyze various problems related to various types of solutions and will be able to prepare various types of solutions.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT I

Elementary Knowledge of Inorganic Chemistry: Structure of atom, atomic weight, molecular and equivalent weight. Acids, bases and salts. Acid Base Balance, pH, pH indicators pH-measurement (pH meter, pH paper, universal indicator). Normality, Molarity and Molality, Mole, Molar solutions, normal solutions, buffer solutions, percent solutions, saturated solutions, standard solutions

Radio isotopes and their use in Biochemistry

UNIT II

Elementary Knowledge of Organic Chemistry: Organic compounds, aliphatic, aromatic, alcohol, ethers, phenols, acids, etc.

Elementary Knowledge of Physical Chemistry- Osmosis, osmotic pressure, dialysis, surface tension, diffusion, hypotonic, hypertonic and isotonic solutions. Definition and classification of some colloids and crystalloids.

UNIT III

Elementary Knowledge of Analytical Chemistry: balances- monopan, twopan, toppan, centrifuges, pH meter, colorimeter, spectrophotometer, fluorimeter, flame photometer, ion selective electrodes, urinometer, chromatograph, electrophoresis, densitometer.

Basic Steps of Analytic Techniques titrimetry, photometry, Electrochemistry, Immuno - chemistry

UNIT IV

Introduction and properties of carbohydrates, proteins and fat.
Introduction and general properties of Nucleic acids and Enzymes
Elementary knowledge of Minerals, Electrolytes and hormones
Therapeutic Drug Monitoring
Metabolism of carbohydrates, proteins and fat.

Practical Work:

1. Preparation and storage of distilled water
2. Preparation of laboratory reagents and standard solutions, storage of chemicals.
3. Units of measurements. S.I. Units, measurement of volume, volumetric apparatus. (Pipettes,
4. flasks, Cylinders)
5. Preparation and standardization of volumetric solutions
6. Preparation of buffer solution and measurement of their pH
7. To prepare the different concentration of solutions
8. To prepare of the 1/10 N HCL
9. To find out the normality of given solution

12. Suggested Resources for Reading

1. A guidebook to Biochemistry Michael Yudkin
2. A Manual of Laboratory & Diagnostic Tests (6/ e) Frances Fischbach
3. Biochemistry Voet and Voet
4. Biochemistry Stryer
5. Biochemistry U. Satyanarayan. & U. Chakrapani
6. Clinical Biochemistry Richard Luxton
7. Clinical Diagnosis & Management by Laboratory method0 (20/e) John Bernard Henary
8. Clinical Biochemistry G. Guru
9. Handbook of Biochemistry M.A. Siddique
10. Textbook of Medical Biochemistry S. Ramkrishnan

11. Biochemical Techniques K. Choudhary
12. Text book of Medical Biochemistry Chaterjee & Shinde
13. Principles of Biochemistry David L. Nelson
14. Textbook of Biochemistry and Human Biology G.P. Talwar
15. Textbook of Medical Laboratory Technology Godkar and Godkar
16. Outline of Biochemistry Conn Stumpf
17. Principles of Internal Medicine Isselbacher
18. Proteins and Proteomics: Laboratory Manual Richard J. Simpson
19. Purifying Proteins for Proteomics: Laboratory Manual Richard J. Simpson
20. Enzymes: Biochemistry, Biotechnology & Clinical chemistry, (2001) Palmer Trevor, Publisher: Horwood Pub. Co., England.
21. Outlines of Biochemistry: 5th Edition, Erice Conn & Paul Stumpf; John Wiley and Sons, USA
22. Fundamentals of Biochemistry. 3rd Edition (2008), Donald Voet & Judith Voet, John Wiley and Sons, Inc. USA
23. Lehninger, Principles of Biochemistry. 5th Edition (2008), David Nelson & Michael Cox, W.H. Freeman and company, NY.
24. Biochemistry: 7th Edition, (2012), Jeremy Berg, Lubert Stryer, W.H. Freeman and company, NY
25. Biochemical Methods for Agricultural Sciences – Sadasivam and Manikam. Wiley Eastern Limited, 1992.
26. Practical Clinical Biochemistry Harold Varley, CBS; 6 edition (1 December 2006)
27. An Introduction to Practical Biochemistry (3rd Edition) – David T Plummer. Tata McGraw-Hill Publishing Company Limited, 1992.
28. Biometrics Identity by Sameer Nanawati
29. Medicinal Chemistry by Ashutosh Kar

Note: Latest edition of text books may be used.

	COURSE TITLE	ANALYTICAL LABORATORY TESTING PROCESS-I
1.	**Course Code**	MLT 203
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100 Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Medical Lab Technician
7.	**Course Objectives**	To gain broad understanding of chemicals/reagents useful in sample analysis. To get the Knowledge of routine hematological tests from urine, stool, semen and sputum tests.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Apply systematized problem-solving techniques to identify and correct procedural errors, identify instrument malfunctions and seek proper supervisory assistance, and verify the accuracy of laboratory results obtained. CO2. Get broad understanding of chemicals/reagents useful in sample analysis. CO3. Operate and maintain laboratory equipment, utilizing appropriate quality control and safety procedure CO4. Well versed with the basic knowledge and handling of chemical/reagents doing analysis (tests like hematological tests, urine, stool, semen and sputum tests). CO5. To offer the assistance to doctors with clinical laboratory tests, performing the vital role in diagnosis of various clinical conditions.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT I

Collection and recording of biological specimens, Collection, Handling and transportation of Blood Sample, Correct method of blood sample collection, Collection methods of samples other than blood samples.
Examinations of sample for serum Urea and Uric acid.
Determination of blood sugar level of plasma (or serum) (a) Orthotoluidine method, (b) Glucose oxidase method Determination od the serum urea nitrogen (a) Diacetyl monoxime method Determination of serum creatinine: Alkaline picrate method Determination of serum total cholesterol

UNIT II

Determination of serum bilirubin a. Malloy and Evelyn b.DMSO method
Determination of TSH Laboratory investigation for megaloblastic anaemia
Platelet function tests and their interpretation
Estimation of Essential electrolytes: Sodium, potassium, calcium, chloride and phosphorus etc.

UNIT III

Erythrocyte sedimentation rate, factors influencing ESR and various procedures for its estimation with their significance. Haematocrit value by macro and micro methods their merits and demerits. Determination of pack cell volume (PCV) Physiological variations in HB, PCV, TLC and platelets
Determination and calculation of red blood indices MCH, MCH, MCHC
Determination of absolute Eosinophil count and Reticulocyte count
Enumeration of formed elements, Automated systems in hematology

UNIT IV

Urine Examination

- Collection, Handling, transportation of Urine
- Exam ination of Urine.
- Urine analysis, routine examination of urine (physical examination of urine) rapid chemical tests of Urine
- Determination of specific gravity of urine by urinometer and refractormeter
- Microscopic examination of urine
- Clinical significance and Detailed Examination of Urine
- Biochemical Test Profile (Quantitative determination of Urine)
- Amylase, Calcium, Chlorides, Creatinine, Sodium, Potassium, Glucose, Proteins, Urea nitrogen, uric acid bile pigments, ketone bodies, porphobilinogen, faecal occult blood.

Practical Work:

All Biochemical tests mentioned in Theory

12. Suggested Resources for Reading

1. Textbook of Medical Laboratory Technology Godkar and Godkar
2. Research Methodology in Medical Sciences Chandorkar
3. Practical Clinical Biochemistry Harold Varley
4. Medical Laboratory Sciences, Theory & Practical A. Kolhatkar
5. Medical Laboratory Technology – Volume I Kanai Mukherjee
6. Medical Laboratory Technology – Volume II Kanai Mukherjee
7. Medical Laboratory Technology – Volume II Kanai Mukherjee
8. Medical Laboratory Technology Methods & Interpretation (5/e) Ramnik Sood
9. Enzymes: Biochemistry, Biotechnology & Clinical chemistry, (2001) Palmer Trevor, Publisher: Horwood Pub. Co. England. Outlines of Biochemistry: 5th Edition, Erice Conn & Paul Stumpf; John Wiley and Sons, USA

10. Fundamentals of Biochemistry. 3rd Edition (2008), Donald Voet & Judith Voet , John Wiley and Sons, Inc. USA
11. Lehninger, Principles of Biochemistry. 5th Edition (2008), David Nelson & Michael Cox, W.H. Freeman and company, NY.
12. Biochemistry: 7th Edition, (2012), Jeremy Berg, Lubert Stryer, W.H.Freeman and company, NY
13. Biochemical Methods for Agricultural Sciences – Sadasivam and Manikam. Wiley Eastern Limited, 1992..
14. Practical Clinical Biochemistry Harold Varley, CBS; 6 edition (1 December 2006)
15. An Introduction to Practical Biochemistry (3rd Edition) – David T Plummer. Tata McGraw-Hill Publishing Company Limited, 1992

Note: Latest edition of text books may be used.